Math 550, S21 **Name:** Answer

Quiz 9

<u>**Instructions:**</u> **There are totally 3 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (6 points) Evaluate $\int_C x^2y\,dx - xy\,dy$, where C is the curve with equation $y^2 = x^3$, from $(1,-1)$ to $(1,1)$.

$$C: \begin{cases} x(t) = t^2, \\ y(t) = t^3, \end{cases} \quad -1 \le t \le 1 \quad \Rightarrow \quad \begin{array}{l} dx = 2t\,dt \\ dy = 3t^2\,dt \end{array}$$

$$\int_C x^2y\,dx - xy\,dy = \int_{-1}^{1} (t^2)^2t^3(2t\,dt) - t^2t^3(3t^2\,dt)$$

$$= \int_{-1}^{1} 2t^8 - 3t^7\,dt$$

$$= \left[\frac{2}{9}t^9 - \frac{3}{8}t^8\right]_{t=-1}^{t=1}$$

$$= \left(\frac{2}{9} - \frac{3}{8}\right) - \left(-\frac{2}{9} - \frac{3}{8}\right)$$

$$= \frac{4}{9}$$

Problem 2. (6 points) Use **Green's theorem** to find the area of the region D enclosed by the hypocycloid $\mathbf{x}(t) = (2\cos^3 t,\, 2\sin^3 t),\ 0 \le t \le 2\pi$.

$$\begin{cases} x(t) = 2\cos^3 t \\ y(t) = 2\sin^3 t \end{cases} \Rightarrow \begin{array}{l} dx = 6\cos^2 t(-\sin t)\,dt, \\ dy = 6\sin^2 t\cos t\,dt, \end{array} \quad 0 \le t \le 2\pi$$

$$\text{Area}(D) = \frac{1}{2}\oint_{\partial D} -y\,dx + x\,dy$$

$$= \frac{1}{2}\int_0^{2\pi} (-2\sin^3 t)(-6\cos^2 t\sin t)\,dt + (2\cos^3 t)(6\sin^2 t\cos t)\,dt$$

$$= 6\int_0^{2\pi} \sin^4 t\cos^2 t + \sin^2 t\cos^4 t\,dt$$

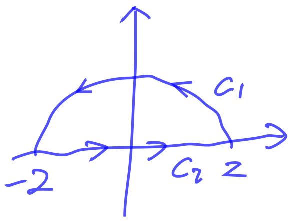
$$= 6\int_0^{2\pi} \sin^2 t\cos^2 t(\sin^2 t + \cos^2 t)\,dt$$

$$= 6\int_0^{2\pi} \sin^2 t\cos^2 t\,dt = \frac{3}{2}\int_0^{2\pi} \sin^2 2t\,dt$$

$$= \frac{3}{2}\int_0^{2\pi} \frac{1-\cos 4t}{2}\,dt = \frac{3}{4}\left[t - \frac{1}{4}\sin 4t\right]_{t=0}^{t=2\pi}$$

$$= \frac{3}{4}\cdot\left[2\pi - 0\right] = \frac{3}{2}\pi$$

Problem 3. (8 points) Verify **Green's theorem** for the vector field $\mathbf{F}(x, y) = 2y\mathbf{i} + x\mathbf{j}$ and the region $D = \{(x, y) \mid x^2 + y^2 \leq 4, y \geq 0\}$.



$\partial D = C_1 \cup C_2$

$C_1 : \begin{cases} x(t) = 2\cos t, \\ y(t) = 2\sin t, \end{cases} \quad 0 \leq t \leq \pi.$

$M(x,y) = 2y$

$N(x,y) = x.$

$C_2 : \begin{cases} x(t) = t, \\ y(t) = 0, \end{cases} \quad -2 \leq t \leq 2$

$$\oint_{\partial D} M\,dx + N\,dy = \iint_D \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\,dA \quad \leftarrow \text{Green's theorem}$$

$D = \{(r, \theta) \mid 0 \leq r \leq 2,\ 0 \leq \theta \leq \pi\}$

$$\iint_D \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\,dA = \iint_D 1 - 2\,dA = \iint_D -dA = -\text{Area}(D)$$

$$= -\frac{1}{2}\cdot\pi\cdot 2^2 = -2\pi$$

$$\oint_{\partial D} M\,dx + N\,dy = \int_{C_1} M\,dx + N\,dy + \int_{C_2} M\,dx + N\,dy$$

$$= \int_{C_1} 2y\,dx + x\,dy + \int_{C_2} 2y\,dx + x\,dy$$

$$= \int_0^{\pi} 2(2\sin t)(-2\sin t)\,dt + 2\cos t(2\cos t)\,dt$$

$$+ \int_{-2}^{2} 0\,dt + 0\,dt$$

$$= \int_0^{\pi} -8\sin^2 t + 4\cos^2 t\,dt$$

$$= \int_0^{\pi} (-8)\cdot\frac{1-\cos 2t}{2} + 4\cdot\frac{1+\cos 2t}{2}\,dt$$

$$= \int_0^{\pi} -2 + 6\cos 2t\,dt$$

$$= -2\pi$$